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An investigation of cultural influences in survivors of pediatric cancer: a systematic review protocol

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Title: An investigation of cultural influences in survivors of pediatric cancer: a systematic review protocol

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ABSTRACT

Introduction: Pediatric cancer affects children and families from diverse backgrounds. However, there is a limited understanding of how diversity/cultural factors play a role, especially in survivorship. This protocol outlines a systematic review on the cultural influences in survivors of childhood cancer. **Methods and Analysis**: This protocol is reported based on the PRISMA-P guidelines checklist and is registered with PROSPERO. EMBASE, MEDLINE, and PsycINFO are searched. Eligibility criteria include original research studies published in English, and an assessment of culture on survivors of childhood cancer. Search terms are developed with a Medical Librarian. Primary outcome will be culture (ethnic and population groups, migration status, acculturation, cultural characteristics) on survivorship and secondary outcome will be psychosocial factors. Data extraction will include participant characteristics such as the number of participants and/or controls, sex, age at diagnosis. Collection will also include analytical approaches, type of cultural variables (predictor, moderator, mediator, outcome), and effect measures. **Ethics and Dissemination:** Ethical approval was not required for this systematic review review. Results from this systematic review will be disseminated in line with PRISMA guidelines through peer-reviewed publications and conference presentations. Findings will also

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be shared with our target communities, survivors of childhood cancer and their families, through the creation of lay summaries and/or educational workshops in the community. Knowledge gathered from this review may help to identify gaps in knowledge and directions for future research. They may also inform the development of clinical recommendations for health care providers of survivors of childhood cancer.

PROSPERO registration number: CRD42021234101.

Strengths and Limitations of the Study

- This systematic review addresses a gap in the current literature on the impact of culture on survivorship in pediatric cancer, including late effects of treatment, follow-up care, psychosocial outcomes, and quality of life.
- This systematic review protocol is registered with PROSPERO and follows the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guidelines with the goal to enhance transparency and replicability of this work, and encourage open science practices.
- This protocol is developed in collaboration with the expertise of a medical librarian to ensure the development of a robust search strategy.
- Exclusion criteria precludes studies that are not published in English. This limitation may result in a biased sample of the literature.

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- Culture is a complex and dynamic construct that may contribute to challenges in the data screening and extraction process, as well as an inconsistent quality of reporting of this construct in the literature.

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INTRODUCTION

With the advancement of pediatric cancer treatment, there is a rapidly growing population of long-term survivors of childhood cancer. This calls to attention the need to better understand the experience of all survivors and ways to inform their long-term follow-up care. Unfortunately, research on the cancer experience of survivors of childhood cancer with consideration of their cultural context is quite limited. In a literature review conducted by Gray and colleagues (2014), the authors examined cultural factors in clinical care and family management for children and their families in various aspects of their cancer experience, ranging from diagnosis, active treatment, to cure/palliative care [1]. The authors concluded from their review that the ways in which cultural factors influence survivorship is poorly understood. Existing studies conducted on the role of culture in the care of survivors have focused on specific ethnic and population groups, including those from Korea [2, 3] and Taiwan [4], as well as Latino youths in the United States (e.g., [5]), regarding their health-related quality of life. These studies uniquely offer a perspective of the lived experience of survivors from selected cultural groups. However, given the scarcity of existing research on this topic, little is known about how culture affects survivorship across a wider range of cultures and geographic regions. For

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example, for survivors of pediatric cancer and their families that live in Canada, issues of acculturation and migration status may be of relevance to their cancer experience and follow-up care. As it stands, research on the influence of culture on survivors of childhood cancer across cultural groups and countries has largely been neglected.

Broadly, the notion of culture can refer to specific groups of individuals who exist in a shared context, including their general characterstics, ideas, beliefs, technology, habits, or practices [6]. As such, studies on culture can vary widely but may include race, ethnicity, and socioeconomic status, among many others. The recently published psychosocial standards of care for childhood cancer determined that poverty is a risk factor that contributes to on-going and escalating stress for survivors and their families, and is strongly influenced by the cultural context [7]. Specifically, low SES coupled with other contextual factors (e.g., age at diagnosis) have been found to be associated with a greater degree of challenges in survivorship [8]. In the same review, ethnic differences were also observed in the cancer experience of survivors. For instance, based on an ethnically diverse sample of adolescents and young adults survivors of pediatric cancer, Arpawong (2013) found that predictors of post-traumatic growth were Hispanic survivors who primarily spoke English at home (vs. Hispanics who primarily spoke Spanish at home) and non-Hispanics [9].

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Taken together, there is some evidence to suggest that cultural factors play a critical role in the late effects and quality of life for survivors of childhood cancer. However, there is a lack of a clear and complete understanding of how cultural factors fully influence pediatric survivorship. Further, to the best of our knowledge, a theoretical framework that accounts for the cultural context in pediatric survivorship currently does not exist. Prior research has largely focused on the effects of family resources (SES, income) and less is known regarding other cultural/diversity factors, as well as their intersection, in pediatric oncology and especially for survivors. The majority of the work on survivors of childhood cancer are either based on Eastern countries or Hispanic immigrants to the United States. Gaining an understanding of the experience of survivor immigrants from other minority groups to North America may be fruitful to our understanding of how acculturation can impact families in their understanding and engagement in treatment, and how such experiences may vary across different ethnic groups in North America. On a global level, knowledge of how survivorship differs across different cultures and geographic regions can help to enhance the generalizability of our findings on and care for all survivors around the world.

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Based on these gaps identified in the literature, there is a need to capture the full range of cultural factors that can potentially affect survivors. Broadly, a deeper understanding of the complex and dynamic process in which culture shapes children and families' cancer experience is critical to provide optimal care. More specifically, knowledge of the experience of survivors from diverse backgrounds can help to determine key aspects of their cancer experience that can be impacted by their cultural context. This knowledge can in turn inform health-care providers in their delivery of culturally-informed care to diverse populations. Accordingly, we endeavor to evaluate the evidence of cultural influences on survivors through a systematic review of the literature. On the basis of parsimony, we narrowed the scope of our review to focus on the study of cultural groups and their influences on pediatric survivors' psychosocial outcomes. Our conceptualization of culture in this review is therefore determined by ethnic and population groups, migration status, acculturation levels, and cultural characteristics (beliefs, attitudes, and behaviours).

Study Objectives

The overarching review question is: How does culture influence pediatric cancer survivors? To answer this question, specific objectives are outlined below.

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Objective 1: To describe the studies that investigate cultural influences in pediatric cancer survivors, including study characteristics (e.g., conceptualizations, operationalizations) and current methods used to measure cultural factors in pediatric cancer survivors (e.g., study design, sample size) coming from a systematic review.

Objective 2: To identify the effects and/or interactions of cultural influence on the psychosocial outcomes of pediatric cancer survivors.

METHODS AND ANALYSIS

This protocol is registered with PROSPERO (CRD42021234101) and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) statement [10]. This systematic review started May 2021 with a preliminary search, pilot of the study selection process, and screening of search results against eligibility criteria. As of February 2022, no data extraction has been conducted. The PRISMA-P checklist was completed. Systematic review progress will be updated on PROSPERO, including any changes to the planned systematic review along with a rationale for the changes.

Patient and public involvement

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This systematic review protocol was developed with patient and public involvement. Specifically, the introduction, objectives, and methodology of this systematic review was developed with feedback provided by our program's patient partners. We will also collaborate with patient partners on the interpretation and dissemination of results.

Eligibility criteria

We will include original research studies that reported the association between cultural influences and psychosocial outcomes of survivors of childhood cancer. Studies must be published in English. Studies that included a wide range of ages and/or a variable time from diagnosis and treatment will be included if the mean age and/or time meet the aforementioned criteria. Studies will include qualitative, quantitative, or mixed method study designs. Narrative review articles and grey literature such as conference and poster abstracts, dissertations, and theses will be excluded. Original study authors will not be contacted directly.

Participants and settings

Children, adolescents, and young adults diagnosed with cancer between 0 and 21 years of age and who are at least 5 years from diagnosis and/or 2 years from the completion of therapy will be considered. Our definition of survivors of childhood cancer is a consistent with the

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definition of survivors established from the Children's Oncology Group guidelines [11].Studies in the context of health, pedatric, psychological, and social/cultural research will be considered. Studies conducted in any geographical locations will be included.

Outcomes

The primary outcome to be reviewed is the study of culture. Culture in this review is defined as the study of cultural groups. Study characteristics will therefore include: ethnic and population groups, migration status, acculturation levels, and cultural norms (beliefs, attitudes, and behaviours). Other culture-related measures used within our scope and conceptualization will be considered. Type of cultural variable considered include predictor, control, moderator, or mediator roles.

Secondary outcomes are measures of psychosocial outcomes characterized according to psychological and social categories. Specifically, psychological outcome variables include (but is not limited to): depression, anxiety, and quality of life. Social outcome variables include (but is not limited to): peer relationships, family relationships, educational and vocational attainment.

Comparator (controls)

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Comparison of intervention/exposure to a control group will be considered but not required. Cross-cultural comparisons, whereas a group of survivors from one country, ethnic, or cultural group, is compared to those of another country, ethnic, or cultural group, will be considered but not required.

Information sources and search strategy

PRISMA guidelines for completion of systematic reviews will be followed. A literature search will be conducted using the following databases: EMBASE, MEDLINE, and PsycINFO. Search terms will be developed with reference to established search filters [12, 13] and in collaboration with a Medical Librarian. A sample of search strategy for OVID Medline can be found in the supplementary information.

Selection process

All exported articles will be subjected to a two-step review process via COVIDENCE [14]. In the first step, all available titles and abstracts will be screened by two independent raters to identify potentially eligible studies for inclusion. Disagreements will be resolved through consensus between these two raters. Inter-rater reliability will be calculated using the Kappa statistic. The identified studies based on the review of titles and abstracts will be retrieved in full

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and again, two independent reviewers will review the content of each full article for eligibility. Disagreements will again be resolved through discussion between raters. Inter-rater reliability will be calculated using the Kappa statistic [15].

Data extraction and management

Retrieved articles will be exported to COVIDENCE, a systematic review management tool to screen title/abstracts and full texts, and extract data. Data abstraction will include the following data: study design type, number of participants, number of controls, sex, median age at diagnosis, age range at diagnosis, median age at evaluation, age range at evaluation, participation rate, median follow-up time, range of follow-up time, outcomes assessment method used and brief summary of study findings. In addition, analytic approach (e.g. regression, tests of interaction between groups, correlations) will be extracted. Type of study cultural variable (predictor, moderator/mediator) will also be extracted. Where possible, effect size and confidence intervals of the effect of predictor (culture) on outcome variables (psychosocial) will be extracted to determine the effect measure for my main outcome variables.

Quality assessment

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Quality assessment will be conducted within each study by considering four primary criteria: 1) selection/subject bias; 2) instrumentation; 3) missing data; and 4) reporting measurement outcomes. The quality of evidence across studies will also be evaluated using the Grading of Recommendations Assessment, Development, and Evaluation framework [16].

Data analysis and synthesis

To address our first objective, studies will be summarized using percentages based on the following criteria: conceptualization of culturally-relevant variables, operationalization of culturally-relevant variables (i.e., measurement tool used); study design, sample size, characteristics of pediatric survivor group studied; and reporter derived from the data abstraction form described above. These findings will be described qualitatively using a narrative synthesis.

To address our second objective, all studies that explore culture and survivors of pediatric cancer will be evaluated to determine if they are associated with any psychosocial outcomes. Data abstraction will adhere to the psychological and social categories as defined in this study. Data extracted abstracted on the effect measures will be summarized quantitatively. Currently, there is no childhood cancer specific model that accounts for the influence of cultural factors on their psychosocial outcomes. As such, evaluation of the associations between cultural variables

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and survivors' psychosocial outcomes will be guided by an existing cultural/acculturation framework; e.g., [17].

ETHICS AND DISSEMINATION

This protocol outlines a systematic review focused on examining the cultural influences in survivors of pediatric cancer. This review does not require ethical approval. Upon completion of the systeatmic review, details of the final publication and/or preprints will be documented and displayed on the public record on PROSPERO. Findings will be shared with academic and clinical stakeholders through traditional academic outputs, including open-access journal publications and national and international academic conferences. Findings will be also shared with our target communities, including survivors of childhood cancer, in collaboration with patient partners, by creating lay summaries, educational workshops, and/or online blogs.

Limitations

This systematic review protocol has limitations worth noting. Culture encompasses a wider range of factors that is not limited to the current variables examined. Other diversity factors such as religion, socio-economic status, sex, gender, and ability level are also important factors to consider in the context of survivors' psychosocial outcomes. Future research on these latter factors can further contribute to our understanding of diversity/cultural factors in play for survivors of pediatric cancer. Another limitation of the current review is our inclusion of original research studies published in English only. This criteria precludes a review of studies conducted

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in other languages that may be fruitful to our research question and therefore limit the generalizability of our findings.

Implications

This review holds both research and clinical implications. First, the current review will advance our understanding of survivorship from diverse populations, with a focus on examining the influence of their cultural context/background on their psychosocial outcomes. Outcomes of this work will in turn help to increase generalizability of research in pediatric oncology with diverse populations, as well as guide future studies in this area. This review may also inform the development and delivery of concrete and culturally specific clinical recommendations for survivors of childhood cancer (see [1] for example), which can in turn address health disparities among underrepresented groups. Ultimatley, a deeper understanding of the complex and dynamic process in which culture shapes children and families' cancer experience is critical to provide optimal care.

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Statements and Declarations

Author Contributions

All authors contributed to the study conception and design, including the literature search strategy. The first draft of the protocol was written by Sharon HJ Hou. Subsequent revisions and edits were provided by Fiona Schulte. All authors read and approved the final manuscript.

Funding Statement

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Competing Interests

The authors have no competing interests to disclose.

Data Availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Ethics Approval

No ethical approval is required for the study protocol of this review.

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to Systematic Reviews from Table 3 in Moher D et al:

*Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Systematic Reviews 2015 **4**:1*

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	33
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	3 – 11
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	282 – 285
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	286 – 291
Sponsor	5b	Provide name for the review funder and/or sponsor	☐	☒	
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	☒	
INTRODUCTION					

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	49 – 109
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	110 – 118
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	133 – 147
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	164 – 169
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	Suppl. Materials
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	179 – 189
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	170 – 178
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	195 – 208
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	148 – 163
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	148 – 163
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	190 – 194
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	195 – 208

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	195 – 208
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐	☒	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐	☒	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☐	☒	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	190 – 194

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Supplementary Information

Sample search strategy for a systematic review of cultural influences in survivors of pediatric cancer

1. Cancer Survivor/ OR exp Neoplasms/ and Survivors/ OR ((cancer* or neoplasm* or tumo?r* or carcinoma*) adj2 survivor*).tw,kf.

2. exp child/ or exp "congenital, hereditary, and neonatal diseases and abnormalities"/ or exp infant/ or adolescent/ or exp pediatrics/ or child, abandoned/ or exp child, exceptional/ or child, orphaned/ or child, unwanted/ or minors/ or (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.

3. Culture/ OR cultur*.tw,kf.OR exp "cultural diversity"/ OR human migration/ or "emigration and immigration"/ OR exp "Emigrants and Immigrants"/ or Refugees/ or (immigrant* or immigration or emigrant* or emigration or refugee* or asylum seeker* or asylee* or displaced person* or "incomer*" or "in comer*" or "new comer*" or newcomer* or migrant* or resettler*).mp. OR acculturation/ OR accult*.tw,kf. OR exp Ethnic Groups/ OR ethnic*.tw,kf.OR exp Minority Groups/ OR exp Minority Health/ OR Continental Population Groups/ OR cultural characteristics/

4. #1 AND #2 AND #3

English language filter applied; date filter up to April 2021 was applied.

Subject heading and key words search.

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Primary Subject Heading:	Oncology
Secondary Subject Heading:	Paediatrics
Keywords:	ONCOLOGY, Paediatric oncology < PAEDIATRICS, Child & adolescent psychiatry < PSYCHIATRY

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INTRODUCTION

With the advancement of pediatric cancer treatment, there is a rapidly growing population of long-term survivors of childhood cancer. This calls to attention the need to better understand the experience of all survivors and ways to inform their long-term follow-up care. Unfortunately, research on the cancer experience of survivors of childhood cancer with consideration of their cultural context is quite limited. In a literature review conducted by Gray and colleagues (2014), the authors examined cultural factors in clinical care and family management for children and their families in various aspects of their cancer experience, ranging from diagnosis, active treatment, to cure/palliative care [1]. The authors concluded from their review that the ways in which cultural factors influence survivorship is poorly understood. Existing studies conducted on the role of culture in the care of survivors have focused on specific ethnic and population groups, including those from Korea [2, 3] and Taiwan [4], as well as Latino youths in the United States (e.g., [5]), regarding their health-related quality of life. These studies uniquely offer a perspective of the lived experience of survivors from selected cultural groups. However, given the scarcity of existing research on this topic, little is known about how culture affects survivorship across a wider range of cultures and geographic regions. For

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example, for survivors of pediatric cancer and their families that live in Canada, issues of acculturation and migration status may be of relevance to their cancer experience and follow-up care. As it stands, research on the influence of culture on survivors of childhood cancer across cultural groups and countries has largely been neglected.

Broadly, the notion of culture can refer to specific groups of individuals who exist in a shared context, including their general characterstics, ideas, beliefs, technology, habits, or practices [6]. As such, studies on culture can vary widely but may include race, ethnicity, and socioeconomic status, among many others. The recently published psychosocial standards of care for childhood cancer determined that poverty is a risk factor that contributes to on-going and escalating stress for survivors and their families, and is strongly influenced by the cultural context [7]. Specifically, low SES coupled with other contextual factors (e.g., age at diagnosis) have been found to be associated with a greater degree of challenges in survivorship [8]. In the same review, ethnic differences were also observed in the cancer experience of survivors. For instance, based on an ethnically diverse sample of adolescents and young adults survivors of pediatric cancer, Arpawong (2013) found that predictors of post-traumatic growth were Hispanic survivors who primarily spoke English at home (vs. Hispanics who primarily spoke Spanish at home) and non-Hispanics [9].

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Taken together, there is some evidence to suggest that cultural factors play a critical role in the late effects and quality of life for survivors of childhood cancer. However, there is a lack of a clear and complete understanding of how cultural factors fully influence pediatric survivorship. Further, to the best of our knowledge, a theoretical framework that accounts for the cultural context in pediatric survivorship currently does not exist. Prior research has largely focused on the effects of family resources (SES, income) and less is known regarding other cultural/diversity factors, as well as their intersection, in pediatric oncology and especially for survivors. The majority of the work on survivors of childhood cancer are either based on Eastern countries or Hispanic immigrants to the United States. Gaining an understanding of the experience of survivor immigrants from other minority groups to North America may be fruitful to our understanding of how acculturation can impact families in their understanding and engagement in treatment, and how such experiences may vary across different ethnic groups in North America. On a global level, knowledge of how survivorship differs across different cultures and geographic regions can help to enhance the generalizability of our findings on and care for all survivors around the world.

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Based on these gaps identified in the literature, there is a need to capture the full range of cultural factors that can potentially affect survivors. Broadly, a deeper understanding of the complex and dynamic process in which culture shapes children and families' cancer experience is critical to provide optimal care. More specifically, knowledge of the experience of survivors from diverse backgrounds can help to determine key aspects of their cancer experience that can be impacted by their cultural context. This knowledge can in turn inform health-care providers in their delivery of culturally-informed care to diverse populations. Accordingly, we endeavor to evaluate the evidence of cultural influences on survivors through a systematic review of the literature. On the basis of parsimony, we narrowed the scope of our review to focus on the study of cultural groups and their influences on pediatric survivors' clinical, psychological, and social outcomes. Accordingly, our conceptualization of culture in this review is determined by ethnic and population groups, migration status, acculturation levels, and cultural characteristics (beliefs, attitudes, and behaviours). Our conceptualization of survivors' outcomes include their clinical outcomes (e.g., treatment engagement and outcomes, late effects, follow up care), psychological outcomes (e.g., anxiety, depression), and social outcomes (e.g., educational and vocational attainment).

Study Objectives

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The overarching review question is: How does culture influence pediatric cancer survivors? To answer this question, specific objectives are outlined below.

Objective 1: To describe the studies that investigate cultural influences in pediatric cancer survivors, including study characteristics (e.g., conceptualizations, operationalizations) and current methods used to measure cultural factors in pediatric cancer survivors (e.g., study design, sample size) coming from a systematic review.

Objective 2: To identify the effects and/or interactions of cultural influence on the outcomes of pediatric cancer survivors, including psychological, social, and clinical outcomes.

METHODS AND ANALYSIS

This protocol is registered with PROSPERO (CRD42021234101) and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) statement [10]. This systematic review started May 2021 with a preliminary search, pilot of the study selection process, and screening of search results against eligibility criteria. As of February 2022, no data extraction has been conducted. The PRISMA-P checklist was completed. Systematic review progress will be updated on PROSPERO, including any changes to the planned systematic review along with a rationale for the changes.

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Patient and Public Involvement

This systematic review protocol was developed with patient and public involvement. Specifically, the introduction, objectives, and methodology of this systematic review was developed with feedback provided by our program's patient partners. We will also collaborate with patient partners on the interpretation and dissemination of results.

Literature Search/Inclusion Criteria

We will include original research studies that reported the association between cultural influences and outcomes (clinical, psychological, social) of survivors of childhood cancer. Studies must be published in English. Studies that included a wide range of ages and/or a variable time from diagnosis and treatment will be included if the mean age and/or time meet the aforementioned criteria. Studies will include quantitative study designs. Narrative review articles and grey literature such as conference and poster abstracts, dissertations, and theses will be excluded. Original study authors will not be contacted directly.

Children, adolescents, and young adults diagnosed with cancer between 0 and 21 years of age and who are at least 5 years from diagnosis and/or 2 years from the completion of therapy will be considered. Our definition of survivors of childhood cancer is a consistent with the

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definition of survivors established from the Children's Oncology Group guidelines [11]. Studies in the context of health, pedatric, psychological, and social/cultural research will be considered. Studies conducted in any geographical locations will be included. No publication date filter was implemented to capture the full range of studies conducted on this subject.

Objectives

The primary objective is to describe culture in pediatric cancer survivors. Culture in this review is defined as the study of cultural groups. Study characteristics will therefore include: ethnic and population groups, migration status, acculturation levels, and cultural norms (beliefs, attitudes, and behaviours). Other culture-related measures used within our scope and conceptualization will be considered. Type of cultural variables considered include predictor, control, moderator, or mediator roles.

The secondary objective is to describe the effects and interactions of culture on clinical, psychological, and social outcomes. Clinical outcome variables include (but is not limited to): late effects, follow-up care, and treatment outcomes. Psychological and social outcomes will be characterized according to psychological and social categories. Specifically, psychological outcome variables include (but is not limited to): depression, anxiety, and quality of life. Social

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outcome variables include (but is not limited to): peer relationships, family relationships, educational and vocational attainment.

Comparator (controls)

Comparison of intervention/exposure to a control group will be considered but not required. Cross-cultural comparisons, whereas a group of survivors from one country, ethnic, or cultural group, is compared to those of another country, ethnic, or cultural group, will be considered but not required.

Information Sources and Search strategy

PRISMA guidelines for completion of systematic reviews will be followed. A literature search will be conducted using the following databases: EMBASE, MEDLINE, and PsycINFO. Search terms will be developed with reference to established search filters [12, 13] and in collaboration with a Medical Librarian. A sample of search strategy for OVID Medline can be found in the supplementary information.

Study Selection

All exported articles will be subjected to a two-step review process via COVIDENCE [14]. In the first step, all available titles and abstracts will be screened by two independent raters

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to identify potentially eligible studies for inclusion. Disagreements will be resolved through consensus between these two raters. Inter-rater reliability will be calculated using the Kappa statistic. The identified studies based on the review of titles and abstracts will be retrieved in full and again, two independent reviewers will review the content of each full article for eligibility. Disagreements will again be resolved through discussion between raters. Inter-rater reliability will be calculated using the Kappa statistic [15].

Data Extraction and Management

Retrieved articles will be exported to COVIDENCE, a systematic review management tool to screen title/abstracts and full texts, and extract data. Data abstraction will include the following data: study design type, number of participants, number of controls, sex, median age at diagnosis, age range at diagnosis, median age at evaluation, age range at evaluation, participation rate, median follow-up time, range of follow-up time, outcomes assessment method used and brief summary of study findings. In addition, analytic approach (e.g. regression, tests of interaction between groups, correlations) will be extracted. Type of study cultural variable (predictor, moderator/mediator) and country in which the study was conducted will be extracted. Where possible, effect size and confidence intervals of the effect of predictor (culture) on

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outcome variables (clinical, psychosocial) will be extracted to determine the effect measure for my main outcome variables.

Quality Assessment

Quality assessment will be conducted within each study by considering four primary criteria: 1) selection/subject bias; 2) instrumentation; 3) missing data; and 4) reporting measurement outcomes. The quality of evidence across studies will also be evaluated using the Grading of Recommendations Assessment, Development, and Evaluation framework [16].

Data analysis and synthesis

To address our first objective, studies will be summarized using percentages based on the following criteria: conceptualization of culturally-relevant variables, operationalization of culturally-relevant variables (i.e., measurement tool used); study design, sample size, characteristics of pediatric survivor group studied; and reporter derived from the data abstraction form described above. These findings will be described qualitatively using a narrative synthesis.

To address our second objective, all studies that explore culture and survivors of pediatric cancer will be evaluated to determine if they are associated with any clinical and/or psychosocial outcomes. Data extraction will adhere to the clinical, psychological, and social categories as

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defined in this study. Data extracted abstracted on the effect measures will be summarized quantitatively. Currently, there is no childhood cancer specific model that accounts for the influence of cultural factors on their outcomes. As such, evaluation of the associations between cultural variables and survivors' outcomes will be guided by an existing cultural/acculturation framework; e.g., [17].

ETHICS AND DISSEMINATION

This protocol outlines a systematic review focused on examining the cultural influences in survivors of pediatric cancer. This review does not require ethical approval. Upon completion of the systeatmic review, details of the final publication and/or preprints will be documented and displayed on the public record on PROSPERO. Findings will be shared with academic and clinical stakeholders through traditional academic outputs, including open-access journal publications and national and international academic conferences. Findings will be also shared with our target communities, including survivors of childhood cancer, in collaboration with patient partners, by creating lay summaries, educational workshops, and/or online blogs.

Limitations

This systematic review protocol has limitations worth noting. Culture encompasses a wider range of factors that is not limited to the current variables examined. Other diversity

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factors such as religion, socio-economic status, sex, gender, and ability level are also important factors to consider in the context of survivors' outcomes. Future research on these latter factors can further contribute to our understanding of diversity/cultural factors in play for survivors of pediatric cancer. Another limitation of the current review is our inclusion of original research studies published in English only. This criteria precludes a review of studies conducted in other languages that may be fruitful to our research question and therefore limit the generalizability of our findings.

Implications

This review holds both research and clinical implications. First, the current review will advance our understanding of survivorship from diverse populations, with a focus on examining the influence of their cultural context/background on their outcomes. Outcomes of this work will in turn help to increase generalizability of research in pediatric oncology with diverse populations, as well as guide future studies in this area. This review may also inform the development and delivery of concrete and culturally specific clinical recommendations for survivors of childhood cancer (see [1] for example), which can in turn address health disparities among underrepresented groups. Ultimatley, a deeper understanding of the complex and dynamic process in which culture shapes children and families' cancer experience is critical to provide optimal care.

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Statements and Declarations

Author Contributions

All authors contributed to the study conception and design, including the literature search strategy. The first draft of the protocol was written by Sharon HJ Hou. Subsequent revisions and edits were provided by Fiona Schulte. All authors read and approved the final manuscript.

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Competing Interests

The authors have no competing interests to disclose.

Data Availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Ethics Approval

No ethical approval is required for the study protocol of this review.

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Supplementary Information

Search strategy for a systematic review of cultural influences in survivors of pediatric cancer

1. Cancer Survivor/ OR exp Neoplasms/ and Survivors/ OR ((cancer* or neoplasm* or tumo?r* or carcinoma*) adj2 survivor*).tw,kf.

2. exp child/ or exp "congenital, hereditary, and neonatal diseases and abnormalities"/ or exp infant/ or adolescent/ or exp pediatrics/ or child, abandoned/ or exp child, exceptional/ or child, orphaned/ or child, unwanted/ or minors/ or (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.

3. Culture/ OR cultur*.tw,kf.OR exp "cultural diversity"/ OR human migration/ or "emigration and immigration"/ OR exp "Emigrants and Immigrants"/ or Refugees/ or (immigrant* or immigration or emigrant* or emigration or refugee* or asylum seeker* or asylee* or displaced person* or "incomer*" or "in comer*" or "new comer*" or newcomer* or migrant* or resettler*).mp. OR acculturation/ OR accult*.tw,kf. OR exp Ethnic Groups/ OR ethnic*.tw,kf.OR exp Minority Groups/ OR exp Minority Health/ OR Continental Population Groups/ OR cultural characteristics/

4. #1 AND #2 AND #3

English language filter applied; date filter up to April 2021 was applied.

Subject heading and key words search.

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to Systematic Reviews from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Systematic Reviews 2015 4:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	34
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	3 – 11
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	242 – 245
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	247 – 251
Sponsor	5b	Provide name for the review funder and/or sponsor	☐	☒	
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	☒	
INTRODUCTION					

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	46 – 108
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	109 – 117
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	172 – 179
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	165 – 170
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	Suppl. Materials
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	180 – 191
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	171 – 179
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	197 – 210
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	147 – 164
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	147 – 164
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	192 – 196
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	180 – 196

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	195 – 208
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐	☒	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐	☒	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☐	☒	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	190 – 194

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